Greenwich

ROYAL OBSERVATORY, GREENWICH.

TELEGRAPHIC DETERMINATIONS

OF

LONGITUDE

MADE IN THE YEARS 1888 TO 1902.

UNIV. OF CALIFORNIA

UNDER THE DIRECTION OF

SIR W. H. M. CHRISTIE, K.C.B., M.A., F.R.S.,

ASTRONOMER-ROYAL.

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1906.

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Price Fifteen Shillings.

Name of Star.	Catalogue.	Magnitude.	Tabular Right Ascension, 1888·0.	Approximate Declination.	Correction to Ephemerides.									
					Bassot.		Defforges.		Turner.		Lewis.		Adopted Means.	
					Corr.	W.	Corr.	W.	Corr.	W.	Corr.	W.	Corr.	W.
			h m s	° ′	s		s		s		s		s	
μ Ceti	L.	4	2. 38. 53·14	+ 9. 38	+0·08	1	...	...	+0·16	1	...	...	+0·12	2
39 Arietis	B.A.C.	4	41. 14·26	+28. 47	...	...	+0·19	2	+0·20	1	+0·16	2	+0·18	5
41 Arietis	F.	3·7	43. 23·42	+26. 48	+0·07	1	+0·01	1	−0·04	1	−0·02	2	0·00	6
σ Arietis	L.	6	45. 18·53	+14. 37	...	...	−0·01	1	−0·05	2	+0·03	1	−0·02	4
η Eridani	L.	3	50. 57·32	− 9. 21	−0·01	1	...	...	−0·03	1	+0·09	1	+0·02	3
ϵ Arietis	F.	4·6	52. 48·46	+20. 53	−0·08	1	−0·17	1	+0·02	2	+0·09	1	−0·03	5
α Ceti	F.	2·6	56. 25·44	+ 3. 39	+0·04	2	−0·04	1	+0·06	2	+0·10	1	+0·04	6
ρ Persei	B.A.C.	4	2. 58. 0·00	+38. 24	+0·01	2	+0·06	1	−0·10	2	−0·03	1	−0·03	6
β Persei	F.	3	3. 0. 52·87	+40. 31	+0·07	2	+0·04	1	−0·05	2	−0·02	1	+0·01	6
δ Arietis	F.	2·3	5. 13·48	+19. 18	−0·07	2	+0·03	1	−0·06	1	−0·10	1	−0·05	5
ζ Arietis	L.	4·5	8. 27·80	+20. 37	−0·04	2	+0·03	1	−0·02	1	−0·07	1	−0·03	5
o Tauri	L.	4	18. 47·11	+ 8. 38	0·00	2	+0·05	1	...	...	...	...	+0·02	3
ξ Tauri	F.	4	21. 5·92	+ 9. 20	−0·01	2	+0·06	1	...	...	...	...	+0·01	3
f Tauri	L.	4·5	24. 41·34	+12. 33	−0·04	2	0·00	1	+0·02	1	...	...	−0·02	4
ϵ Eridani	F.	3·7	3. 27. 39·27	− 9. 50	+0·01	1	−0·10	1	...	...	...	...	−0·05	2

XIII. CLOCK ERROR.

The Clock Error was determined from observation of the stars given in the preceding section. The following table contains the times of transit of these stars and of the circumpolar stars used to determine the Azimuth Error. Column 1 contains the star's name; column 2, the position of the micrometer head, whether East or West. Column 3 gives the number of wires over which the transit was observed, or, in the case of circumpolar stars, the number of bisections made, as explained on p. 25. Column 4 gives the reading at which the R.A. micrometer was set during the evening for observations of clock stars, and the mean of the micrometer-readings at which bisections were made for the polar stars. The occasions when a polar star was bisected by a wire other than the middle, or when it was observed without moving the R.A. micrometer (*see* p. 24), are mentioned in the notes, the wires being numbered as on p. 10. Column 5 gives, in all cases, the time of transit over the middle wire, set at the reading of the R.A. micrometer adopted for the day, corrected for parallax of pens. Column 6 gives the transit corrected for Collimation, Level, and Azimuth Error (p. 62). Column 7 contains the Tabular Apparent R.A. of the stars taken from the ephemerides before mentioned, corrected by application of the corrections given on p. 63. Column 8 shows the difference of columns 6 and 7, which is the Clock Error indicated by the individual stars. Column 9 is the Clock Error at 0^h local Sidereal time on each day, obtained from each star. The corresponding Clock Errors deduced from polar stars are given in column 8 as a check in the accuracy of the reduction, but are not given in column 9, as they are, of course, not used in determining final Clock Error.

STAR-TRANSITS OBSERVED AT PARIS

AND

RESULTING CLOCK ERRORS.

Star's Name.	Position.	No. of Wires.	R.A. Micrometer.	Observed Transit.	Corrected Transit.	Tabular Apparent R.A.	Clock Apparently Slow.	Clock Apparently Slow at 0^h Sidereal.
			r	h m s	s	s	s	s
1888 September 30. Observer H T. *Transit B.*								
λ Ursæ Minoris	W	10	10·204	19. 35. 5·32	25·82	40·59	14·77	...
Groombridge 1119 S.P.	W	10	10·846	19. 44. 50·17	28·00	37·78	9·78	...
τ Aquilæ	W	9	10·130	19. 58. 32·74	31·80	41·86	10·06	9·72
64 Aquilæ	W	9	,,	20. 2. 7·64	6·64	16·56	9·92	9·58
θ Aquilæ	W	9	,,	20. 5. 24·48	23·48	33·34	9·86	9·53
20 Vulpeculæ	W	9	,,	20. 7. 11·32	10·53	20·52	9·99	9·66
ρ Aquilæ	W	5	,,	20. 8. 58·39	57·51	7·38	9·87	9·54
α^2 Capricorni	W	9	,,	20. 11. 43·60	42·51	52·33	9·82	9·50
β^2 Capricorni	W	7	,,	20. 14. 36·35	35·24	45·04	9·80	9·48
B.A.C. 7014	W	9	,,	20. 17. 30·55	29·60	39·45	9·85	9·54
69 Aquilæ	W	6	,,	20. 23. 40·96	39·95	49·63	9·70	9·40
ε Delphini	W	9	,,	20. 27. 44·58	43·67	53·46	9·79	9·49
ζ Delphini	W	9	,,	20. 29. 57·11	56·22	6·02	9·80	9·50
Bradley 2701	W	4	,,	20. 33. 43·25	44·75	55·52	10·77	...
Bradley 2701	E	4	,,	20. 33. 44·14	45·92	55·52	9·60	...
75 Draconis	E	4	,,	20. 35. 4·90	6·67	17·30	10·63	...
3 Aquarii	E	9	,,	20. 41. 42·66	41·68	51·61	9·93	9·65
15 Delphini	E	6	,,	20. 44. 10·42	9·56	19·38	9·82	9·54
μ Aquarii	E	9	,,	20. 46. 29·87	28·86	38·78	9·92	9·65
32 Vulpeculæ	E	9	,,	20. 49. 39·70	38·98	49·05	10·07	9·80
61^1 Cygni	E	9	,,	21. 1. 45·37	44·76	54·78	10·02	9·76
61^2 Cygni	E	9	,,	21. 1. 47·01	46·40	56·35	9·95	9·69
ν Aquarii	E	9	,,	21. 3. 22·88	21·84	31·70	9·86	9·61
γ Equulei	E	9	,,	21. 4. 46·63	45·76	55·69	9·93	9·68
ζ Cygni	E	6	,,	21. 8. 2·88	2·18	12·14	9·96	9·72
α Equulei	E	9	,,	21. 10. 6·54	5·63	15·55	9·92	9·68
σ Cygni	E	9	,,	21. 12. 53·95	53·35	3·07	9·72	9·49
Groombridge 3548	E	5	11·201	21. 21. 47·55	53·76	2·94	9·18	...
16 Pegasi	W	9	10·130	21. 47. 50·88	50·08	0·15	10·07	9·88
B.A.C. 7641	W	9	,,	21. 51. 21·80	20·90	30·87	9·97	9·79
18 Pegasi	W	7	,,	21. 54. 25·57	24·62	34·41	9·79	9·61
α Aquarii	W	9	10·130	21. 59. 55·34	54·34	4·13	9·79	9·62
ι Pegasi	W	9	,,	22. 1. 41·00	40·20	50·14	9·94	9·77
Bradley 2993	W	10	9·406	22. 22. 8·08	12·10	21·30	9·20	...
10 Lacertæ	E	9	10·130	22. 34. 7·69	7·08	16·74	9·66	9·54
ζ Pegasi	E	9	,,	22. 35. 45·96	45·09	54·93	9·84	9·72
η Pegasi	E	9	,,	22. 37. 38·46	37·76	47·62	9·86	9·74
τ^2 Aquarii	E	9	,,	22. 43. 33·51	32·46	42·10	9·64	9·53
λ Aquarii	E	9	,,	23. 46. 39·89	38·88	48·67	9·79	9·68
34 Piscium	E	9	,,	0. 4. 10·61	9·74	19·39	9·65	9·66
γ Pegasi	E	9	,,	0. 7. 21·86	21·03	30·68	9·65	9·66
35 Piscium	E	9	,,	0. 9. 6·45	5·56	15·19	9·63	9·64
Bradley 1672 S.P.	E	10	10·162	0. 13. 38·59	21·96	30·26	8·30	...
Bradley 1672 S.P.	W	10	10·234	0. 13. 31·24	17·11	30·26	13·15	...
51 Piscium	W	9	10·130	0. 26. 31·17	30·22	39·58	9·36	9·40
13 Ceti	W	9	,,	0. 29. 23·10	22·08	31·48	9·40	9·44
20 Ceti	W	9	,,	0. 47. 11·23	10·23	19·52	9·29	9·36
μ Andromedæ	W	9	,,	0. 50. 26·51	25·84	35·31	9·47	9·55
2 Ursæ Minoris	W	6	10·005	0. 53. 43·68	47·78	56·28	8·50	...
φ Piscium	W	9	,,	1. 7. 34·23	33·42	42·82	9·40	9·49
37 Ceti	W	9	,,	1. 8. 39·71	38·65	47·97	9·32	9·41
ſ Piscium	W	9	,,	1. 11. 55·32	54·35	3·78	9·43	9·53
υ Piscium	W	9	,,	1. 13. 12·82	12·03	21·42	9·39	9·49
Polaris	W	10	10·863	1. 18. 39·91	56·16	59·15	2·99	...
Polaris	E	10	11·113	1. 18. 35·58	53·80	59·15	5·35	...
η Piscium	E	9	11·130	1. 25. 23·15	22·32	32·05	9·73	9·85
101 Piscium	E	9	,,	1. 29. 40·91	40·14	49·71	9·57	9·70
π Piscium	E	9	,,	1. 31. 3·61	2·75	12·26	9·51	9·64
105 Piscium	E	9	,,	1. 33. 32·13	31·30	40·89	9·59	9·73
ν Piscium	E	9	,,	1. 35. 29·96	29·05	38·70	9·65	9·79
1888 October 1. Observer H T. *Transit B.*								
61^1 Cygni	W	9	10·140	21. 1. 46·95	46·68	54·76	8·08	7·78
61^2 Cygni	W	9	,,	21. 1. 48·48	48·21	56·33	8·12	7·82
ν Aquarii	W	9	,,	21. 3. 24·11	23·52	31·69	8·17	7·87
γ Equulei	W	9	,,	21. 4. 48·00	47·54	55·68	8·14	7·85
ζ Cygni	W	7	,,	21. 8. 4·25	3·92	12·13	8·21	7·92
α Equulei	W	9	,,	21. 10. 7·90	7·41	15·54	8·13	7·85
σ Cygni	W	9	,,	21. 12. 55·08	54·82	3·05	8·23	7·95
1 Pegasi	W	9	,,	21. 16. 48·74	48·33	56·45	8·12	7·85
Groombridge 3548	W	6	9·888	21. 21. 49·11	53·40	2·60	9·20	...
Groombridge 3548	E	6	10·733	21. 21. 47·58	53·12	2·60	9·48	...
ε Pegasi	E	3	10·140	21. 38. 35·62	35·23	43·27	8·04	7·80
14 Pegasi	E	9	,,	21. 44. 47·56	47·31	55·61	8·30	8·08
16 Pegasi	E	9	,,	21. 47. 52·32	52·04	60·13	8·09	7·87
B.A.C. 7641	E	2	,,	21. 51. 23·16	22·78	30·86	8·08	7·86
ζ Pegasi	E	9	10·140	22. 35. 47·28	46·90	54·93	8·03	7·89
η Pegasi	E	9	,,	22. 36. 39·76	39·51	47·61	8·10	7·96
λ Pegasi	E	9	,,	22. 41. 3·05	2·75	10·60	7·85	7·72
λ Aquarii	E	5	,,	22. 46. 41·21	40·72	48·67	7·95	7·83
51 Pegasi	E	9	,,	22. 51. 52·62	52·30	0·27	7·97	7·85
3 Piscium	E	6	,,	22. 54. 48·33	47·88	55·70	7·82	7·71
ο Andromedæ	E	9	,,	22. 56. 40·98	40·84	48·92	8·08	7·98
β^1 Pegasi	E	9	,,	22. 58. 15·46	15·19	23·25	8·06	7·96
α Pegasi	E	7	,,	22. 59. 5·76	5·41	13·38	7·97	7·87
55 Pegasi	E	9	,,	23. ½. 16·58	16·19	24·17	7·98	7·88
58 Pegasi	E	9	,,	23. 4. 17·88	17·49	25·48	7·99	7·90
59 Pegasi	E	7	,,	23. 5. 59·63	59·23	7·33	8·10	8·01
φ Aquarii	E	7	,,	23. 8. 26·27	25·78	33·74	7·96	7·88
Bradley 3147	E	10	10·013	23. 28. 0·75	6·61	14·46	7·85	...

September 30. λ Ursæ Minoris W was observed on wire VII. Bradley 2701 E and 75 Draconis E were observed without moving the micrometer. Bradley 1672 S.P. W was observed on wire II.

October 1. Groombridge 3548 W was observed on wire IV. Groombridge 3548 E was observed on wire III. Bradley 3147 E was observed on wire IX.

TRANSITS OBSERVED AT GREENWICH

AND

RESULTING CLOCK ERRORS.

Star's Name.	Position.	No. of Wires.	R.A. Micrometer.	Observed Transit.	Corrected Transit.	Tabular Apparent R.A.	Clock Apparently Slow.	Clock Apparently Slow at 0^h Sidereal.	Star's Name.	Position.	No. of Wires.	R.A. Micrometer.	Observed Transit.	Corrected Transit.	Tabular Apparent R.A.	Clock Apparently Slow.	Clock Apparently Slow at 0^h Sidereal.
1888 September 30. Observer L. *Transit C.*																	
			r	h m s	s	s	s	s				r	h m s	s	s	s	s
3 Aquarii	E	9	9·900	20. 41. 45·67	45·73	51·61	+ 5·88	5·91	35 Piscium	W	9	9·900	0. 9. 9·31	9·22	15·19	5·97	5·97
15 Delphini	E	9	,,	20. 44. 13·41	13·56	19·38	5 82	5·85	36 Piscium	W	9	,,	0. 10. 45·51	45·42	51·31	5·89	5·89
μ Aquarii	E	9	,,	20. 46. 32·82	32·86	38·78	5·92	5·95	Bradley 1672 S.P.(E. & E.)	W	4	,,	0. 13. 26·87	20·24	30·26	10·02	...
32 Vulpeculæ	E	5	,,	20. 49. 43·10	43·36	49·05	5·69	5·72	δ Andromedæ	W	4	,,	0. 33. 17·41	17·44	23·17	5·73	5·73
21 Capricorni	E	7	,,	20. 54. 29·60	29·59	35·55	5·96	5·98	ζ Andromedæ	W	9	,,	0. 41. 21·12	21·11	26·85	5·74	5·73
α Equulei	E	9	,,	21. 10. 9·57	9·68	15·55	+ 5·87	5·99	δ Piscium	W	9	,,	0. 42. 49·08	48·98	54·84	5·86	5·85
Groombridge 3548 (E. & E.)	E	6	,,	21. 21. 45·44	52·29	2·94	+10·65	...									
									υ Piscium	E	9	,,	1. 13. 17·58	15·83	21·42	5·59	5·58
34 Piscium	W	7	,,	0. 4. 13·59	13·51	19·39	5 88	5·88	Polaris (E. & E.)	E	4	,,	1. 18. 31·78	50·14	59·15	9·01	...
γ Pegasi	W	9	,,	0. 7. 25·05	24·99	30·68	5·69	5·69									
1888 October 1. Observer L. *Transit C.*																	
λ Ursæ Minoris...(E. & E.)	E	20	10·275	19. 35. 20·16	26·70	39·19	12·49	...	λ Aquarii	E	9	9·900	22. 46. 42·27	42·60	48·67	6·07	6·08
									δ Aquarii	E	9	,,	22. 48. 38·34	38·67	44·79	6·12	6·13
15 Delphini	W	9	9·900	20. 44. 13·27	13·43	19·37	5·94	5·96	51 Pegasi	E	4	,,	22. 51. 53 92	54·27	0·27	6·00	6·01
μ Aquarii	W	5	,,	20. 46. 32·81	32·95	38·77	5·82	5·84	o Andromedæ	E	9	,,	22. 56. 42·61	43·00	48·92	5·92	5·93
32 Vulpeculæ	W	9	,,	20. 49. 43·09	43·25	49·03	5·78	5·80	β^1 Pegasi	E	9	,,	22. 58. 16·89	17·25	23·25	6·00	6·01
21 Capricorni	W	9	,,	20. 54. 29·55	29·68	35·54	5·86	5·88	α Pegasi	E	5	,,	22. 59. 7·00	7·35	13·38	6·03	6·04
α Equulei	W	9	,,	21. 10. 9·75	9·90	15·54	5·64	5·65	55 Pegasi	E	9	,,	23. 1. 17·83	18·17	24·17	6·00	6·00
σ Cygni	W	9	,,	21. 12. 57·36	57·51	3·05	5·54	5·55	Bradley 3147	E	19	9·511	23. 28. 6·63	8·84	14·46	5·62	...
1 Pegasi	W	9	,,	21. 16. 50·67	50·83	56·45	5·62	5·63									
Groombridge 3548	W	10	9·219	21. 21. 57·52	56·62	2·60	5·98	...	Bradley 3147	W	21	2·841	23. 28. 9·13	8·20	14·46	6·26	...
									ω Piscium	W	9	9·900	23. 53. 29·86	30·01	36·10	6·09	6·09
κ Aquarii	E	9	9·900	22. 31. 53·35	53·68	59·68	6·00	6·01	30 Piscium	W	9	,,	23. 56. 9·09	9·24	15·44	6·20	6·20
10 Lacertæ	E	9	,,	22. 34. 10·42	10·80	16·73	5·93	5·94	2 Ceti	W	9	,,	23. 57. 56·13	56·26	2·61	6·35	6·35
ζ Pegasi	E	9	,,	22. 35. 48·66	49·00	54·93	5·93	5·94	33 Piscium	W	9	,,	23. 59. 32·18	32·33	38·61	6 28	6·28
η Pegasi	E	9	,,	22. 37. 41·37	41·74	47·61	5·87	5·88	5 Ceti	W	9	,,	0. 2. 24·24	24·39	30·46	6·07	6·07
λ Pegasi	E	9	,,	22. 41. 4·32	4·68	10·60	5·92	5·93	34 Piscium	W	9	,,	0. 4. 13·24	13·40	19·39	5·99	5·99
τ^2 Aquarii	E	9	,,	22. 43. 35 80	36·13	42·10	5·97	5·98	γ Pegasi	W	9	,,	0. 7. 24·38	24·54	30·68	6·14	6·14
1888 October 3. Observer L. *Transit C.*																	
θ Aquarii	W	5	9·900	22. 10. 51·41	51·60	57·68	6·08	6·08	γ Pegasi	E	9	9·900	0. 7. 24·39	24·76	30·69	5·93	5·93
γ Aquarii	W	9	,,	22. 15. 48·35	48·52	54·55	6·03	6·03	35 Piscium	E	9	,,	0. 9. 9·10	9·49	15·20	5·71	5·71
Bradley 2993	W	15	9·500	22. 22. 19·20	15·16	20·73	5·57	...	36 Piscium	E	9	,,	0. 10. 45·23	45·62	51·32	5·70	5·70
η Aquarii	W	9	9·900	22. 29. 31·97	32·14	38·33	6·19	6·19	Bradley 1672 S.P.	E	15	8·700	0. 13. 21·18	24·33	30·70	6·37	...
κ Aquarii	W	9	,,	22. 31. 53·91	54·09	59·67	5·58	5·58	51 Piscium	E	9	9·900	0. 26. 33·48	33·88	39·59	5·71	5·71
10 Lacertæ	W	4	,,	22. 34. 10·64	10·61	16·70	6·09	6·09	13 Ceti	E	9	,,	0. 29. 25·44	25·87	31·50	5·63	5·63
ζ Pegasi	W	9	,,	22. 35. 49·12	49·25	54·91	5·66	5·66	π Andromedæ	E	6	,,	0. 30. 50·76	51·06	56·85	5·79	5·79
η Pegasi	W	9	,,	22. 37. 41·88	41·91	47·59	5·68	5·68	δ Andromedæ	E	9	,,	0. 33. 17·18	17·50	23·18	5·68	5·68
λ Pegasi	W	9	,,	22. 41. 4·50	4·57	10·59	6·02	6·02	2 Ursæ Minoris	E	20	2·313	0. 53. 51·24	50·45	56·30	5·85	...
τ^2 Aquarii	W	9	,,	22. 43. 36·07	36·28	42·09	5·81	5·81									
λ Aquarii	W	9	,,	22. 46. 42·65	42·84	48·66	5·82	5·82	2 Ursæ Minoris	W	19	7·442	0. 53. 53·15	48·98	56·30	7·32	...
δ Aquarii	W	9	,,	22. 48. 38·78	38·99	44·78	5·79	5·79	η Ceti	W	4	9·900	1. 2. 53·79	53·99	59·82	5·83	5·83
51 Pegasi	W	9	,,	22. 51. 54·49	54·57	0·26	5·69	5·69	33 Ceti	W	9	,,	1. 4. 44·15	44·31	50·27	5·96	5·96
									Polaris	W	10	11·814	1. 19. 11·70	56·91	59·54	2·63	...
33 Piscium	E	4	,,	23. 59. 32·27	32·71	38·61	5·90	5·90									
34 Piscium	E	9	,,	0. 4. 13·22	13·61	19·39	5·78	5·78	Polaris	E	15	10·176	1. 18. 56·88	53·47	59·54	6·07	...

September 30. Groombridge 3548 E and Bradley 1672 S.P. were observed without moving the micrometer. Polaris was observed on wires VI., VII., VIII., IX.

The means of the four results from each set of signals are given in Column IV. of the following Table. The means of the results "Signals sent from Paris," and "Signals sent from Greenwich," were taken, and the difference of these is given in Column VI.

Resulting Difference of Longitude and Means for each Day.

Date.		Greenwich Sid. Time.	Chrono-graphs.	Resulting Longitude.	Mean for Day.	Time of Transmission.	Mean for Day.
	d	h m		m s	m s	s	s
Sept.	30	21. 54	FF′	9. 20·648		+ 0·028	
		23. 12	FE′	·625		+ 0·045	
		23. 23	EE′	·593		+ 0·043	
		23. 45	EF′	·608	9. 20·619	+ 0·028	+ 0·036
Oct.	1	21. 41	FF′	·570		+ 0·035	
		21. 53	FE′	·585		+ 0·050	
		22. 4	EE′	·530		+ 0·025	
		22. 19	EF′	·535	9. 20·555	+ 0·030	+ 0·035
Oct.	3	23. 8	FF′	·785		+ 0·085	
		23. 22	FE′	·743		+ 0·048	
		23. 33	EE′	·720		+ 0·015	
		23. 44	EF′	·715	9. 20·741	+ 0·020	+ 0·042
Oct.	5	21. 52	FF′	·735		+ 0·025	
		22. 5	FE′	·725		+ 0·020	
		22. 17	EE′	·698		+ 0·043	
		22. 32	EF′	·705		+ 0·020	
		0. 7	FF′	·725		+ 0·030	
		0. 18	FE′	·688		− 0·008	
		0. 31	EE′	·728		+ 0·013	
		0. 42	EF′	·760	9. 20·721	+ 0·020	+ 0·020
Oct.	11	21. 56	FF′	·438		+ 0·033	
		22. 31	FE′	·443		+ 0·038	
		22. 40	EE′	·403		+ 0·013	
		22. 59	EF′	·410		+ 0·035	
		0. 41	FF′	·408		+ 0·048	
		0. 53	EE′	·363	9. 20·411	+ 0·063	+ 0·038
Oct.	13	21. 49	FF′	·428		+ 0·048	
		22. 1	FE′	·438		+ 0·028	
		22. 35	EE′	·503		+ 0·088	
		22. 48	EF′	·430		+ 0·040	
		0. 38	FF′	·365		+ 0·020	
		1. 00	EE′	·393	9. 20·426	+ 0·048	+ 0·045
Oct.	14	22. 43	FF′	·443		+ 0·028	
		23. 40	FE′	·420		+ 0·055	
		23. 59	EE′	·433		+ 0·073	
		0. 34	EF′	·510		+ 0·035	
		0. 57	FF′	·485		+ 0·035	
		1. 11	EE′	·495	9. 20·464	+ 0·035	+ 0·044
Oct.	15	0· 3	FE′	·590	9. 20·590	+ 0·050	+ 0·050
Oct.	17	22. 5	FF′	·583		+ 0·033	
		22. 32	FE′	·523		+ 0·048	
		22. 43	EE′	·513		+ 0·058	
		22. 56	EF′	·550		+ 0·035	

Thus we have the following results :—

	m	s		s
From Parts I. and III.	9	20·777	±	0·025
„ „ II.	9	20·427	±	0·025

Longitude of the Paris instrument East of the Greenwich instrument :—

$$9^m 20^s{\cdot}602 \pm 0^s{\cdot}018$$

This longitude refers to the following meridians :—

At Paris, the pier of the English instrument which is $0^s{\cdot}27$ west of Cassini's meridian.

At Greenwich the old meridian of Bradley, 19 ft. or $0^s{\cdot}02$ west of the Transit-Circle.

Thus the meridian of Cassini is,

$$9^m\ 20^s{\cdot}60 - 0^s{\cdot}02 + 0^s{\cdot}27 \text{ or}$$

$$9^m\ 20^s{\cdot}85 \text{ E.}$$

of the Greenwich Transit Circle.

Transit E. Paris—continued.

Values of n.

Date.	Names of Stars.	Position.	Tabular R.A.—Observed Transit.	Factors.	n Apparent.	n Adopted.	Date.	Names of Stars.	Position.	Tabular R.A.—Observed Transit.	Factors.	n Apparent.	n Adopted.
1892.			s		s	s	1892.			s		s	s
July 21	Groomb. 1119 S.P.	W	+ 9·91	−54·82	+0·303		July 22	λ Ursæ Minoris.....	W	+47·56	+55·67	+0·357	
	β Capricorni.....	W	+26·46	− 0·27				*b* Aquilæ.....	W	+27·73	+ 0·21		
	Groomb. 3548.....	W	+28·97	+16·77	+0·131	0·250		λ Ursæ Minoris....	E	+46·79	+55·67	+0·347	+0·309
	ζ Cygni.....	W	+26·84	+ 0·57				16 Delphini.....	E	+27·62	+ 0·22		
	Groomb. 3548.....	E	+29·58	+16·77	+0·207			Groomb. 1119 S.P.	E	+11·80	−54·82	+0·287	
	μ Capricorni.....	E	+26·06	− 0·25				16 Delphini.....	E	+27·62	+ 0·22		
	Bradley 1399 S.P..	E	+23·86	−10·99	+0·220			Groomb. 3548.....	E	+32·79	+16·77	+0·310	
	η Aquarii.....	E	+26·26	− 0·01				α Equulei.....	E	+27·62	+ 0·08		
July 22	δ Ursæ Minoris.....	W	+32·97	+16·88	+0·309			Groomb. 3548.....	W	+32·80	+16·77	+0·297	
	μ Herculis.....	W	+27·92	+ 0·53				γ Capricorni.....	W	+27·72	− 0·31		
	δ Ursæ Minoris.....	E	+31·73	+16·88	+0·255			Bradley 1399 S.P..	W	+24·72	−10·99	+0·273	
	1 Aquilæ.....	E	+27·39	− 0·15				θ Pegasi.....	W	+27·75	+ 0·10		
	Cephei 51 S.P.....	E	+24·25	−20·55	+0·137			Bradley 1399 S.P..	E	+23·17	−10·99	+0·413	
	29 Sagittarii.....	E	+27·01	− 0·37				θ Pegasi.....	W	+27·75	+ 0·10		
	Cephei 51 S.P.....	W	+19·53	−20·55	+0·409								
	π Sagittarii.....	W	+27·73	− 0·39									

Transit E was moved in Azimuth between June 12 and 13.

Value of the Instrumental Constants l, m and n.

In the following tables the values of l (the level-error) converted into seconds of time, the values of n and the deduced values of m are collected.

The values of m are derived from the adopted values of l (the level-error) and n by the formula,

$$m = (l - n \sin \lambda) \sec. \lambda \text{ where } \lambda \text{ is the latitude.}$$

The adopted values for λ are :—

Latitude of Greenwich, 51° 28′ 38″.

„ „ Montsouris, 48° 49′ 18″.